Exp. No 03 **Date: 23- 07- 2026**

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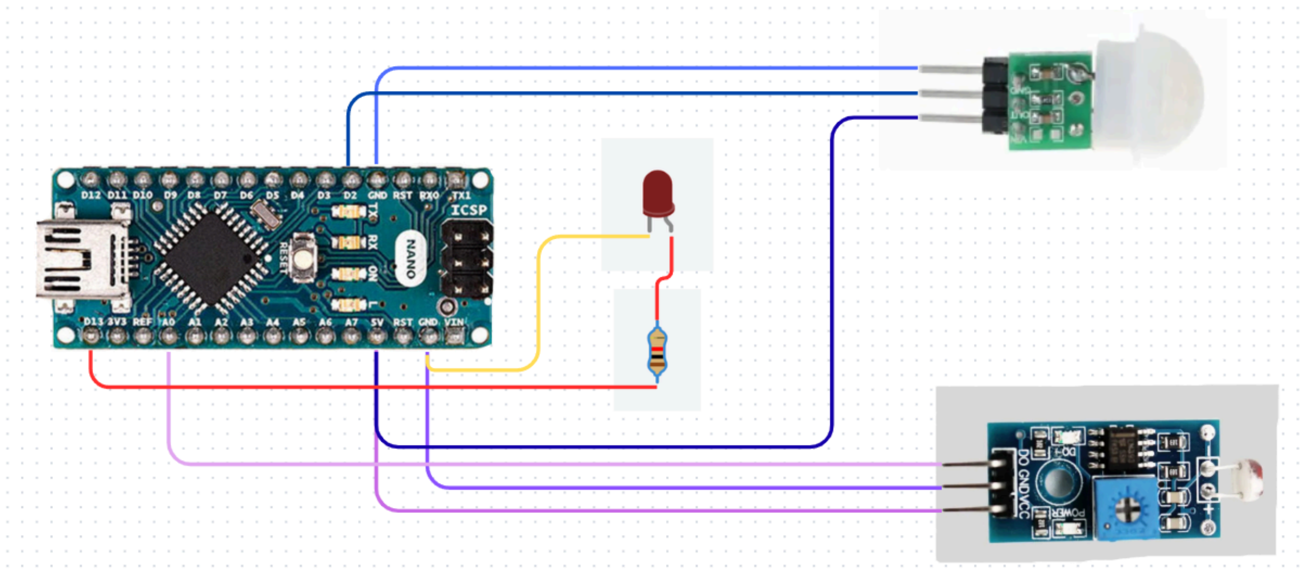
Motion-Activated and Light-Dependent Staircase Lighting System Using PIR and LDR Sensors

● Components Required

Sr.No.	Component	Quantity	Description	Use
1.	Arduino Nano	1	A small microcontroller board used for programming electronic projects.	Used to pass the code of detecting day or night and if motion is present.
2.	Breadboard	1	A board with small holes used to make circuits.	Used to make the circuit easily and test connections.
3.	PIR Motion Sensor	1	A sensor that detects movement of people or objects.	Used to detect motion and turn the light ON.
4.	LDR (Light Dependent Resistor)	1	A sensor that detects light and darkness.	Used to check whether it is day or night.
5.	LED	1	Acts as the light.	Lights in dark when motion captured.
6.	Resistor	1	A resistor is used to reduce the flow of current.	Used to protect the LED from excessive current.

● Circuit Diagram

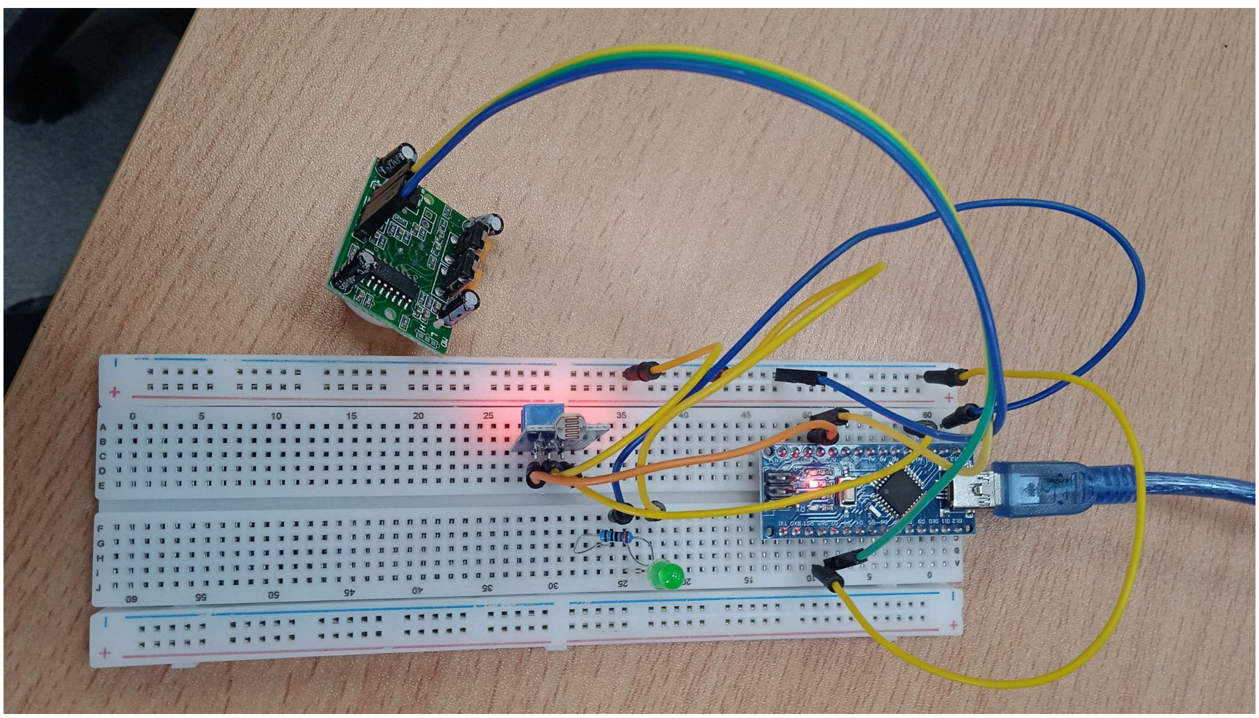


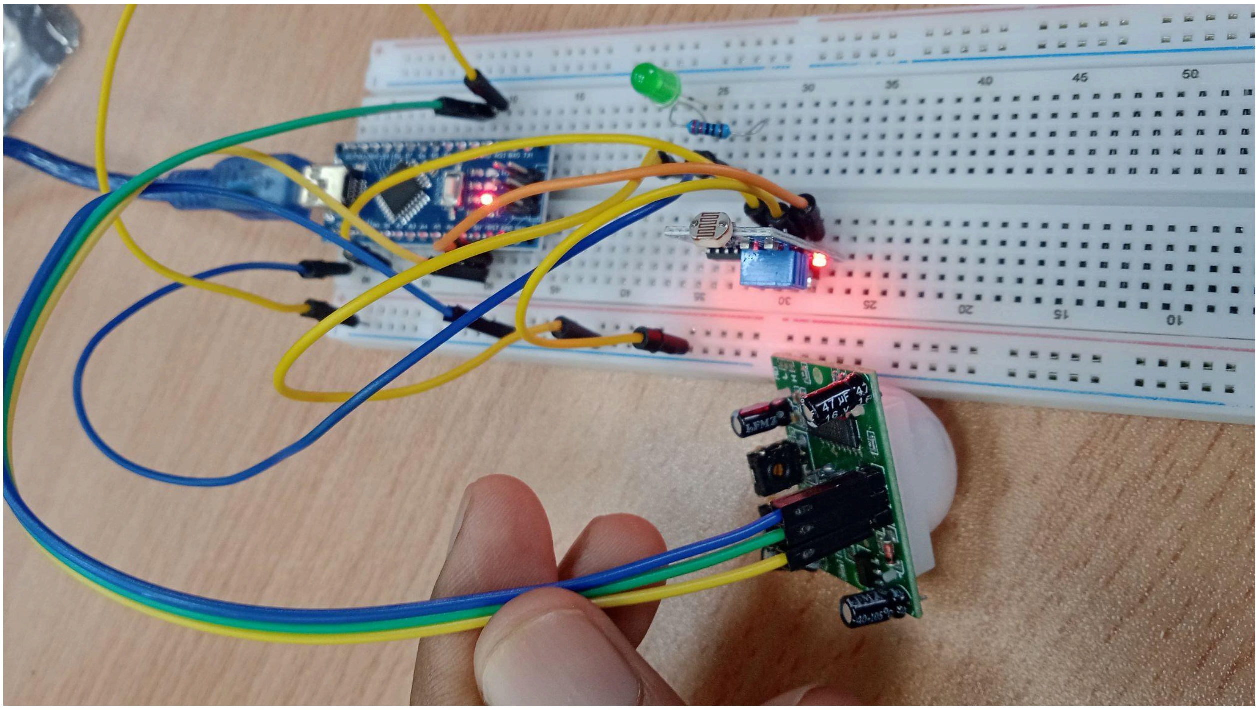
- **Program Code**

```
const int ldrPin = A0;
const int pirPin = 2;
const int ledPin = 13;

// Adjust this after testing your LDR
const int threshold = 500;

void setup() {
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int ldrValue = analogRead(ldrPin);
  int motion = digitalRead(pirPin);

  Serial.print("LDR: ");
  Serial.print(ldrValue);
  Serial.print("  Motion: ");
  Serial.println(motion);

  // Day
  if (ldrValue > threshold) {
     if (motion == HIGH) {
     digitalWrite(ledPin, HIGH);
     delay(2000);
```

```
    } else {
      digitalWrite(ledPin, LOW);
    }
  }
  // Night
  else {
      digitalWrite(ledPin, LOW);

  }

  delay(100);
}
```

- **Photograph of the Experiment**





- **Conclusion**

In this experiment, we learned how to use an LDR and a PIR motion sensor with Arduino. The LDR detects day and night, PIR sensor detects motion. Programmed the Arduino to turn the LED OFF during the day and ON only at night when motion is detected.